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Math 3108 – Linear Algebra
Exam 2
Prof. Runnion
October 18, 2024

Directions:

- 1) All cell phones and other electronic noisemaking devices must be turned completely off (silence or on vibrate is *not* acceptable) and put away for the duration of the exam.
- 2) No calculators, books, or other materials are permitted.
- 3) You must write **darkly** and **legibly** – this exam will be scanned for electronic grading.
- 4) Show **ALL** your work! Correct answers which are not properly justified will not receive full credit.
- 5) Failure to follow directions specific to a problem will result in the loss of points.
- 6) Circle, box, or underline each final answer if necessary for clarity. Label all graphs.
- 7) If you work a problem two different ways, clearly indicate which one you want us to grade, preferably by crossing out the one you do not want us to look at. If you use two methods, one of which is wrong, and neither method is crossed out, you will not receive full credit.
- 8) Answers must be exact (like $\sqrt{2}$), not approximate (like 1.414), unless a problem specifically indicates otherwise.
- 9) Simplify where appropriate. Quantities such as $\sqrt{9}$ and $\cos \pi$ should be calculated.
- 10) Extra space has been provided on all problems. There should be more than sufficient space to work all problems in the space indicated. Thus, NO WORK APPEARING ON THE BACK OF PAGES WILL BE GRADED!
- 11) All work must appear within this exam packet to be graded. **No scratch paper is allowed under any circumstances.**
- 12) This packet has seven sheets of paper, including this cover page. Do NOT remove the staple or remove any sheet from this packet.
- 13) Once this exam begins, you will have 50 minutes to complete your solutions.

**DO NOT OPEN THIS EXAM
UNTIL TOLD TO DO SO**

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Multiple Choice and True/False. Fill in the box corresponding to the correct answer.

1. (3 pts) A basis for a vector space V is a _____ of vectors in V .
☐ i) minimal spanning set ☐ ii) maximal linearly independent set
☒ iii) both (a) and (b) ☐ iv) neither (a) nor (b)
2. (3 pts) For an $n \times n$ matrix A , if the equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution, then the columns of A span $\mathbb{R}^n$.
☒ i) Always true ☐ ii) Sometimes true ☐ iii) Never true
3. (3 pts) If there is an $n \times n$ matrix C such that $CA = I$, then the equation $A\mathbf{x} = \mathbf{b}$ has a solution for each $\mathbf{b}$ in $\mathbb{R}^n$.
☒ i) Always true ☐ ii) Sometimes true ☐ iii) Never true
4. (3 pts) If A and B are invertible $n \times n$ matrices, then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.
☒ i) Always true ☐ ii) Sometimes true ☐ iii) Never true
5. (4 pts) If the rows of the 4×4 matrix A are linearly independent, which of the following is true? Select all that apply.
☐ i) The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
☒ ii) The column space of A is a subspace of $\mathbb{R}^4$.
☒ iii) The rows of A span $\mathbb{R}^4$.
☐ iv) The determinant of A is zero.
6. (4 pts) Let A be an $m \times n$ matrix. Which of the following is a subspace of $\mathbb{R}^m$? Select all that apply.
☒ i) $\text{Col } A$ (the column space of A)
☐ ii) $\text{Nul } A$ (the null space of A)
☐ iii) $\text{Row } A$ (the row space of A)

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Open Answer. Show ALL your work! Place your final answer in the box provided, if applicable.

7. (6 points) Show that the set $H = \left\{ \begin{bmatrix} a+2b \\ 3b \\ a-b \end{bmatrix} : a, b \text{ real} \right\}$ is closed with respect to scalar

multiplication. *Note that this can be done the standard way or a "clever" way, both of which work quite nicely.*

$$\begin{bmatrix} 1 & 2 \\ 0 & 3 \\ 1 & -1 \end{bmatrix} \text{ is linearly independent}$$

8. a) (6 points) Find a matrix A such that $W = \left\{ \begin{bmatrix} 5c-4d \\ 2c+d \\ c \end{bmatrix} : c, d \text{ real} \right\} = \text{Col } A$.

$$\begin{bmatrix} 5 & -4 \\ 2 & 1 \\ 1 & 0 \end{bmatrix}$$

Solution: $A = \begin{bmatrix} 5 & -4 \\ 2 & 1 \\ 1 & 0 \end{bmatrix}$

- b) (2 points) The set W is a subspace of. (Fill in the box.)

☐ i) $\mathbb{R}^2$ ☒ ii) $\mathbb{R}^3$ ☐ iii) Neither of these.

9. The matrices A and B given below are row equivalent. (You do *not* need to check this.)

$$A = \begin{bmatrix} 2 & -4 & -1 & -3 & 5 & 2 \\ -1 & 2 & -4 & -3 & -7 & -1 \\ -1 & 2 & -13 & -12 & -16 & -1 \\ 5 & -10 & 5 & -10 & 20 & 5 \end{bmatrix}$$

$$B = \begin{bmatrix} \boxed{1} & -2 & 0 & 0 & 3 & 1 \\ 0 & 0 & \boxed{1} & 0 & 1 & 0 \\ 0 & 0 & 0 & \boxed{1} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- a) (4 points) Find a basis for the column space of A .

$$\text{basis Col } A = \left\{ \begin{bmatrix} 2 \\ -1 \\ -1 \\ 5 \end{bmatrix}, \begin{bmatrix} -1 \\ -4 \\ -13 \\ 5 \end{bmatrix}, \begin{bmatrix} -3 \\ -3 \\ -12 \\ -10 \end{bmatrix} \right\}$$

- b) (10 points) Find a basis for the null space of A .

$$x_1 = 2x_2 - 3x_5 - x_6$$

$$x_3 = -x_5$$

$$x_4 = 0$$

$$x_2, x_5, x_6 \text{ are free}$$

$$\vec{x} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_6 \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{basis Nul } A = \left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

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10. (10 points) Consider the system of equations
$$\begin{cases} x + 2z = 3 \\ -2x + y + z = 4 \\ x - 3y + z = 1 \end{cases}$$

Given that $\begin{vmatrix} 1 & 0 & 2 \\ -2 & 1 & 1 \\ 1 & -3 & 1 \end{vmatrix} = 14$, use Cramer's Rule to solve for the variable y only.

$$y = \frac{\begin{vmatrix} 1 & 3 & 2 \\ -2 & 4 & 1 \\ 1 & 1 & 1 \end{vmatrix}}{14} = \frac{0}{14} = 0$$

$$\begin{vmatrix} 1 & 3 & 2 \\ -2 & 4 & 1 \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 3 & 2 \\ 0 & 10 & 5 \\ 0 & -2 & -1 \end{vmatrix} \xrightarrow{\frac{1}{5}R_2 + R_3 \rightarrow R_3} \begin{vmatrix} 1 & 3 & 2 \\ 0 & 10 & 5 \\ 0 & 0 & 0 \end{vmatrix} = 1(10)(0) = 0$$

Solution: $y =$

0

11. (8 points) Assume that the matrices shown below are partitioned conformably for block multiplication. Compute the product.

$$\begin{bmatrix} I & 0 \\ -X & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} IA+0 & IB+0 \\ -XA+IC & -XB+ID \end{bmatrix}$$

Solution:

$$\begin{bmatrix} A & B \\ -XA+IC & -XB+ID \end{bmatrix}$$

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12. (12 points) Find the LU factorization of the matrix $A = \begin{bmatrix} 4 & 3 & -5 \\ -4 & -5 & 7 \\ 8 & 6 & -8 \end{bmatrix}$.

$$A = \begin{bmatrix} 4 & 3 & -5 \\ -4 & -5 & 7 \\ 8 & 6 & -8 \end{bmatrix} \xrightarrow{\substack{R_1 + R_2 \rightarrow R_2 \\ -2R_1 + R_3 \rightarrow R_3}} \begin{bmatrix} 4 & 3 & -5 \\ 0 & -2 & 2 \\ 0 & 0 & 2 \end{bmatrix} = U$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

$$\text{Solution: } L = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 4 & 3 & -5 \\ 0 & -2 & 2 \\ 0 & 0 & 2 \end{bmatrix}$$

13. (6 points) Show that the set of all polynomials of the form $\mathbf{p}(t) = a + bt$ where a and b are real is closed with respect to vector addition.

$$\mathbf{p}(t) = 0 + 0t = 0$$

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14. a) (6 points) Use the method of your choice to calculate A^{-1} given $A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$.

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 3 & 2 & 0 & 1 \end{array} \right] \xrightarrow{-\frac{3}{2}R_1 + R_2 \rightarrow R_2} \left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 0 & \frac{1}{2} & -\frac{3}{2} & 1 \end{array} \right] \xrightarrow{2R_2} \left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 0 & 1 & -3 & 1 \end{array} \right] \xrightarrow{-R_2 + R_1 \rightarrow R_1} \left[\begin{array}{cc|cc} 2 & 0 & 4 & -1 \\ 0 & 1 & -3 & 1 \end{array} \right]$$

$$\xrightarrow{\frac{1}{2}R_1} \left[\begin{array}{cc|cc} 1 & 0 & 2 & -\frac{1}{2} \\ 0 & 1 & -3 & 1 \end{array} \right]$$

Solution: $A^{-1} = \begin{bmatrix} 2 & -\frac{1}{2} \\ -3 & 1 \end{bmatrix}$

- b) (4 points) Use the inverse you found in part (a) to solve $A\mathbf{x} = \mathbf{b}$ given $\mathbf{b} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$.

$$\mathbf{x} = A^{-1}\mathbf{b} \quad \begin{bmatrix} 2 & -\frac{1}{2} \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 7 \\ -11 \end{bmatrix}$$

Solution: $\mathbf{x} = \begin{bmatrix} 7 \\ -11 \end{bmatrix}$

15. (6 points) Find the area of the parallelogram whose vertices are $(0,0)$, $(-2,3)$, $(3,3)$, and $(5,0)$. Place your answer in the box provided.

$$A = \begin{bmatrix} -2 & 3 \\ 3 & 3 \end{bmatrix} \quad |\det A| = |-6 - 9| = 15$$

Area = 15